Mineralogy, Petrology and Mineral chemistry of NWA 8173: An anomalous enstatite achondrite with evidence for high temperature sulphidation

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Summary

Amongst enstatite rich meteorites are a small subset of achondritic rocks that vary texturally and mineralogically from the Aubrites [1,3]. Extensive metamorphism of an enstatite chondrite precursor [3] may be responsible for forming these anomalous enstatite achondrites and their formation mechanism may have been related to endogenous heating (via decay of short-lived nuclides such as ^{26}Al) or impact heating. In this study we report and describe the mineralogy, chemistry and petrography of the NWA 8173 anomalous enstatite achondrite and provide insight about the metamorphic history of this meteorite. Several unique features that make NWA 8173 distinct from other enstatite achondrites are described here. Our findings indicate NWA 8173 formed from a high temperature melt followed by annealing, exsolution and local remobilization of sulfide phases. This meteorite may represent extreme sulfidation of a EH-like precursor.

Findings

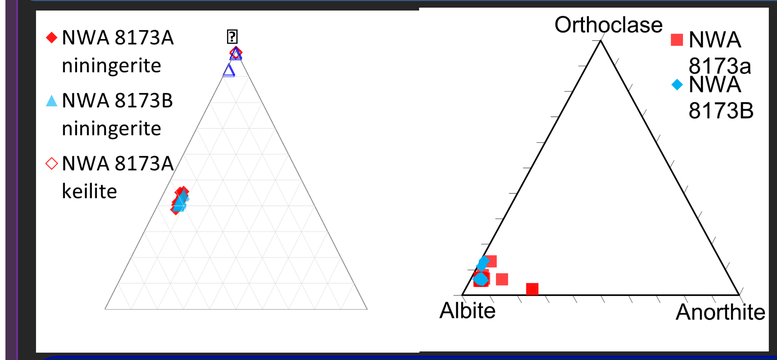


Fig 1a) Sulfide ternary for NWA 8173a-b. b) Feldspar ternary compositions of NWA 8173a-b.

Unlike most enstatite meteorites, Niningerite-keilite is the dominant sulfide in NWA 8173, with daubréelite as the principal exsolved lamellae. Oldhamite and trace amounts of troilite were also observed. Higher amounts of chromium and from 0.3-0.4% in Cr-poor sulfides.(what does this sentence fragment mean?) Exsolution types on Fig 2(a-d) show daubréelite, keilite and oldhamite as the exsolved phases in niningerite . Schreibersite present has a composition of -(14-15% P), Si (0.3-0.43%), Ni 5%, and Co 0.3-0.36%.

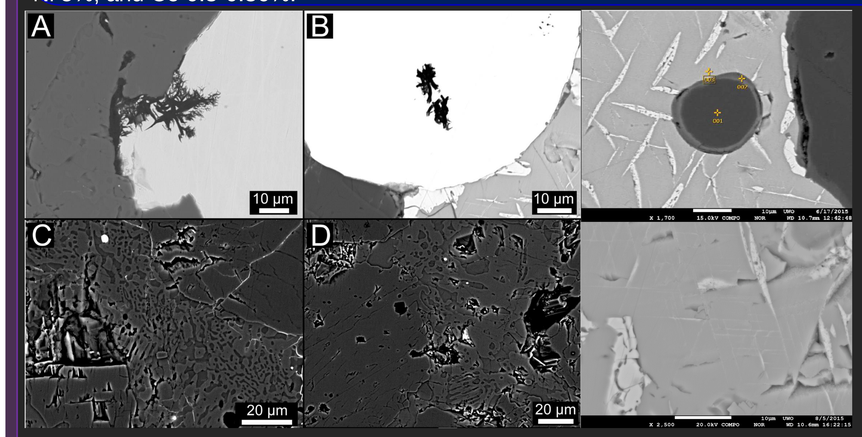
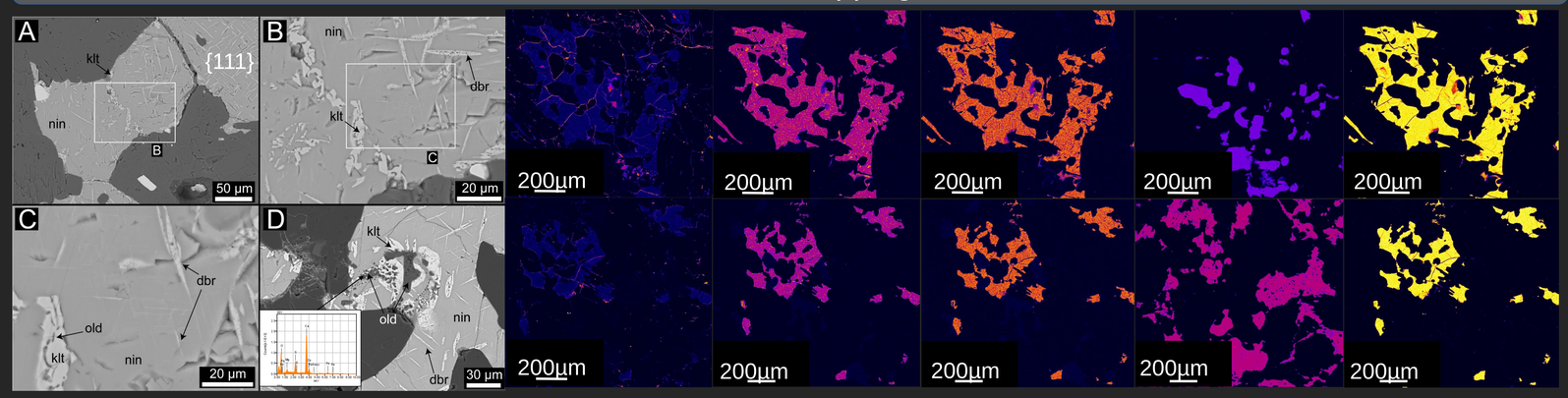


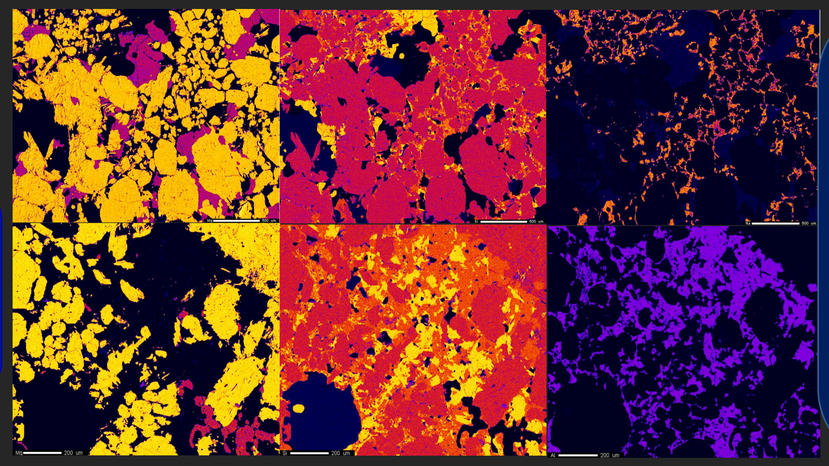
Fig 2.BSE image of dendritic graphite inclusion in metals grains of NWA 8173a-b. b) Micrographic texture in interstitial plagioclase and SiO_2 grains of NWA 8173a-b. e) SiO_2 included in sulfides f) Daubreelite exsolution at micro scale

Methods

Textural relationships in two polished samples of contrasting lithology were examined using reflected light microscopy and electron backscatter imaging.
•A Jeol JXA-8530f field emission electron microprobe was used to obtain quantitative mineral compositions, using a 15 kV and 20nA beam current.
•*In situ* mineral ID was obtained using a Bruker D8 Discover µXRD at Western University [4], operating with Co Kα radiation (λ = 1.78897 Å) at 35 kV and 45 mA, using a 60 mm Göbel mirror and nominal incident beam diameter of 300 µm.

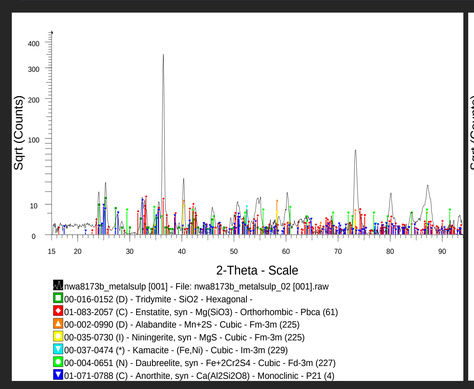
Elemental mapping

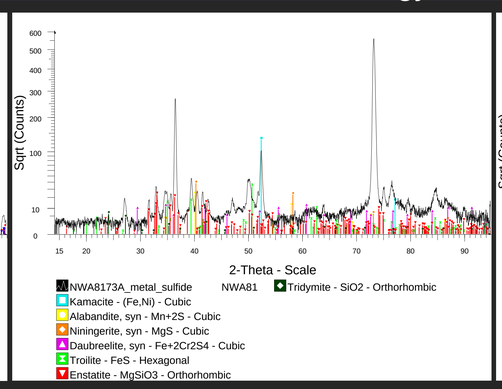


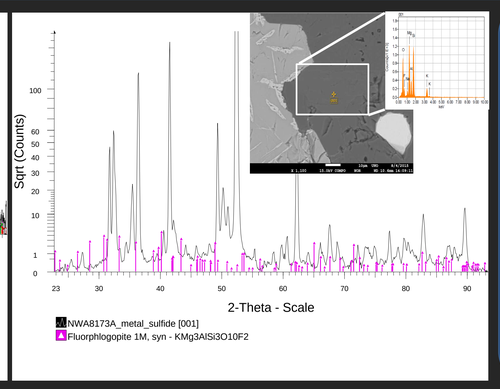


Plagioclase included in metal (Fig 3k,l) and interstitial micrographic silica-plagioclase intergrowths (Fig2c,d) may be representative of two feldspar formation episodes, with the latter forming by localized melting in a late stage heating event, followed by annealing producing the micrographic intergrowth. . The paucity of troilite, predominance of niningerite-keilite, chemical purity of enstatite (Fs 0.01-0.16, Ws 0.23-0.31) all support a role for the sulfidation of ferromagnesian silicates during formation of NWA8173, this reaction has been described for EH3 chondrites [6]. The high silicon content of NWA 8173 in metal suggests a role for the disproportionation of SiO (as a component of silicates) into Si and SiO2 at high temperature under very reducing conditions. Localized melting may have produced smaller, sub-rounded grains, found interstitially along plagioclase and SiO_2. High temperature heating is supported by the presence of tridymite (860-1470 ºC [7](**still need to add this reference**) (Fig 4a,b).

Mineralogy







Future work:
- ^{26}Al/^{26}Mg isotopic study to determine possible heat source.
- Sulfide thermometry
- Shock assessment via µXRD

References:
[1] Boesenberg et al. 2014 LPSC XLV abstract #1486, [2] Keil & Bishoff, 2008 *MAPS* 43, 1233-1240, NWA 2526 [3] Izawa et al, 2011 MAPS 46, 1742-1753, [4] Flemming, 2007 *Can. J. Earth Sci.* 44, 1333-1346, [5] Fleet & MacRae 1986 *GCA 51 1511-1521,* [6] *Lehner et al, 2012 LPSC XLIII abstract #2309.*

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